

Analysis of Acoustic Features and TTS Naturalness

Which acoustic features are associated with TTS naturalness?



Automata **Final Project Report**

Acoustic-TTS-Analysis

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Abstract

AI-generated speech from Text-to-Speech (TTS) systems is increasingly judged on how natural it sounds to human listeners, typically measured with a Mean Opinion Score (MOS). This project investigates which measurable acoustic characteristics - pitch, energy, spectral shape, and silence patterns - are associated with perceived naturalness. Using the VoiceMOS Challenge dataset of TTS audio samples and their MOS ratings, acoustic features were extracted with the librosa library from 3,382 audio clips, merged with naturalness scores, and analyzed through correlation analysis and linear regression. Spectral contrast and silence ratio showed the strongest individual associations with MOS, while spectral rolloff and spectral centroid carried the greatest weight in a joint linear model. However, the model explained only about 18% of the variance in MOS scores, suggesting that simple acoustic summary statistics capture only part of what makes synthetic speech sound natural to human listeners.

1. Introduction

Text-to-Speech systems have improved rapidly, but not all synthetic voices sound equally natural. Understanding which acoustic properties drive human perception of naturalness can help guide TTS system design and evaluation. This project uses a data-driven approach: extracting quantitative acoustic features from TTS audio samples and statistically relating them to human-assigned naturalness ratings (MOS).

1.1 Research Questions

- Main RQ: Which acoustic features are associated with TTS naturalness?
- RQ1: How are naturalness (MOS) scores distributed across the TTS systems in the dataset?
- RQ2: Which acoustic features show the strongest relationship with TTS naturalness?
- RQ3: Do TTS systems with higher naturalness scores exhibit different acoustic characteristics than lower-scoring systems?

1.2 Dataset

The project uses the VoiceMOS Challenge dataset, a publicly available collection of TTS audio samples paired with human-assigned Mean Opinion Score (naturalness) ratings, obtained from Zenodo. The dataset provides 3,382 matched audio-clip / MOS-rating pairs across multiple TTS systems, with no duplicate ratings per file.

2. Methodology

The analysis pipeline was implemented in Python across three Jupyter notebooks, moving from raw data to final statistical model:

Notebook	Purpose
01_explore_dataset.ipynb	Load and validate MOS and feature tables; check for duplicates and join integrity
02_feature_extraction.ipynb	Extract acoustic features from raw .wav files using librosa; store results in SQLite
03_analysis.ipynb	Merge features with MOS; run EDA, correlation analysis, and linear regression

2.1 Tools and Technologies

- Python, Pandas, NumPy: data handling
- Librosa: acoustic feature extraction
- SQLite3: structured storage of features and MOS labels
- Matplotlib, Seaborn: visualization
- scikit-learn: regression modeling and evaluation

2.2 Data Preparation

Audio files and their MOS labels were verified to be free of duplicates and fully join-compatible on the file name key. All 3,382 audio files were then processed through a feature-extraction pipeline; a `wav_analysis` table was created in SQLite as an inner join between the extracted acoustic features and the MOS scores, forming the analysis dataset.

2.3 Acoustic Feature Extraction

For each audio file, the following features were computed using librosa:

- Duration: total clip length in seconds
- Pitch statistics: mean, standard deviation, min, max, and range, derived from YIN pitch tracking (50–400 Hz)
- Energy (RMS) statistics: mean, standard deviation, and range
- Spectral features: spectral rolloff, spectral centroid, spectral bandwidth, and spectral contrast
- Silence ratio: proportion of the clip identified as non-speech, based on amplitude-based voice activity splitting
- Zero-crossing rate (ZCR): a measure of signal noisiness/frication

2.4 Statistical Analysis

Pearson correlation coefficients were computed between each acoustic feature and MOS to identify individual linear relationships. A multiple linear regression model was then trained (80/20 train-test split, standardized features) to assess the joint predictive contribution of all features together, evaluated using R^2 , Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

3. Results and Findings

3.1 RQ1: Distribution of Naturalness Scores

Figure 1 shows the distribution of MOS scores across all 3,382 rated audio samples. Naturalness ratings span the available MOS range with a roughly bell-shaped, moderately spread distribution, indicating a mix of higher- and lower-naturalness synthetic speech samples in the dataset rather than scores clustered at one extreme.

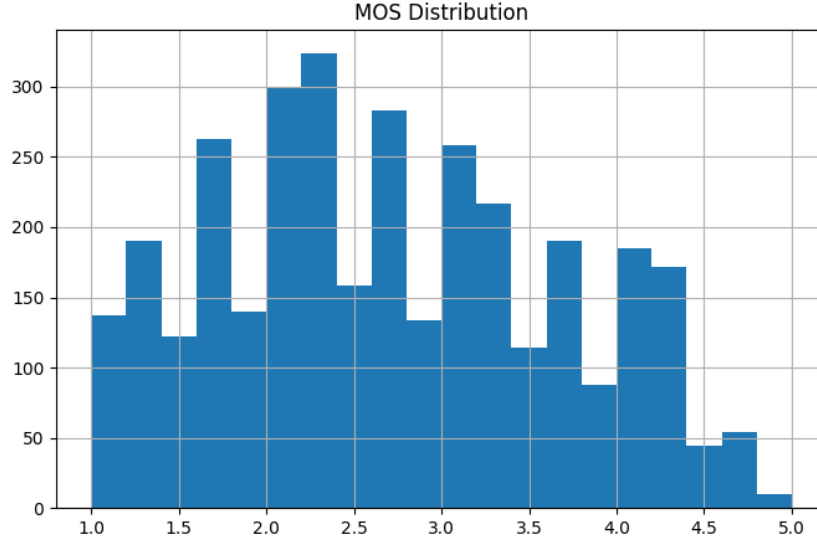


Figure 1. Distribution of MOS naturalness scores across the dataset.

3.2 RQ2: Feature Correlations with Naturalness

Figure 2 shows the full correlation matrix across all extracted acoustic features and MOS. Correlation with MOS specifically, sorted by strength, is summarized in Table 2.

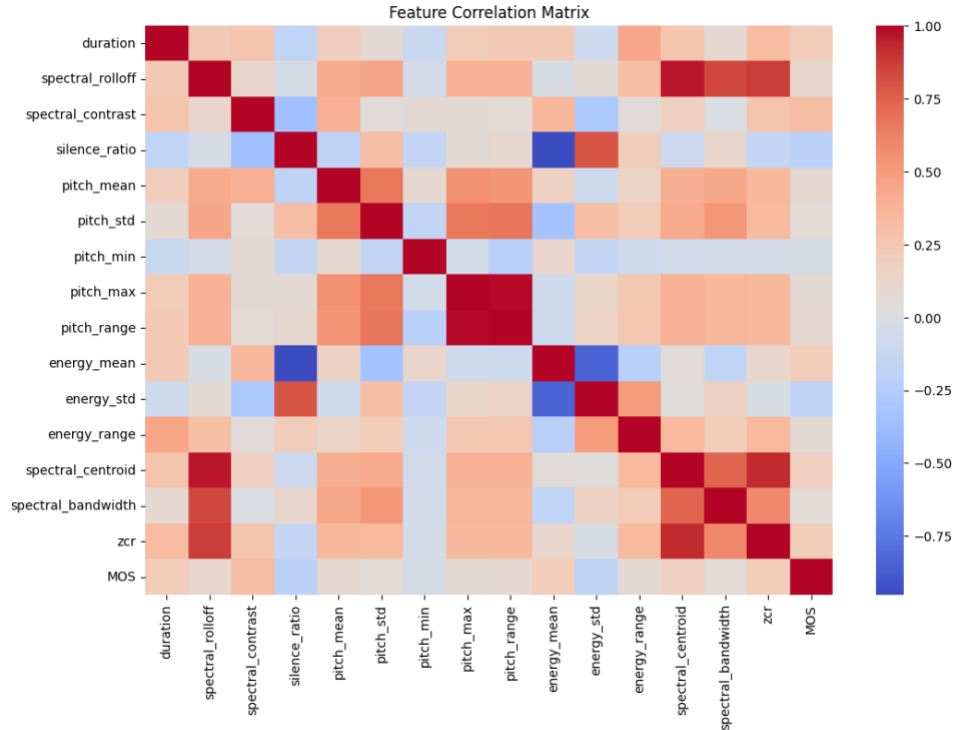


Figure 2. Correlation heatmap across all acoustic features and MOS.

Feature	Correlation with MOS
spectral_contrast	0.313

Feature	Correlation with MOS
zcr	0.212
duration	0.205
energy_mean	0.203
spectral_centroid	0.179
spectral_rolloff	0.112
pitch_mean	0.090
pitch_range	0.088
pitch_max	0.084
energy_range	0.076
pitch_std	0.067
spectral_bandwidth	0.055
pitch_min	− 0.035
energy_std	− 0.185
silence_ratio	− 0.203

Table 2. Pearson correlation of each acoustic feature with MOS. *Table 2. Pearson correlation of each acoustic feature with MOS.*

Spectral contrast shows the strongest positive relationship with naturalness ($r = 0.313$), while silence ratio shows the strongest negative relationship ($r = -0.203$) — clips with proportionally more silence tend to be rated less natural. Figures 3 and 4 visualize these two relationships directly.

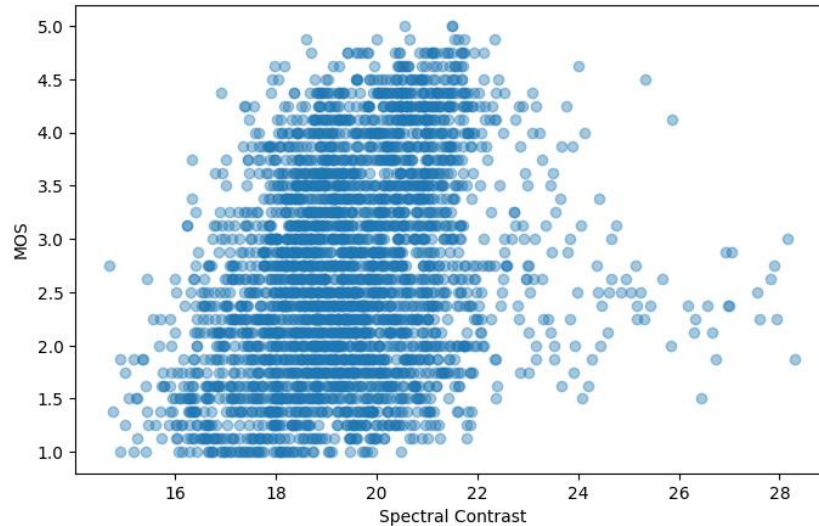


Figure 3. Spectral contrast vs. MOS.

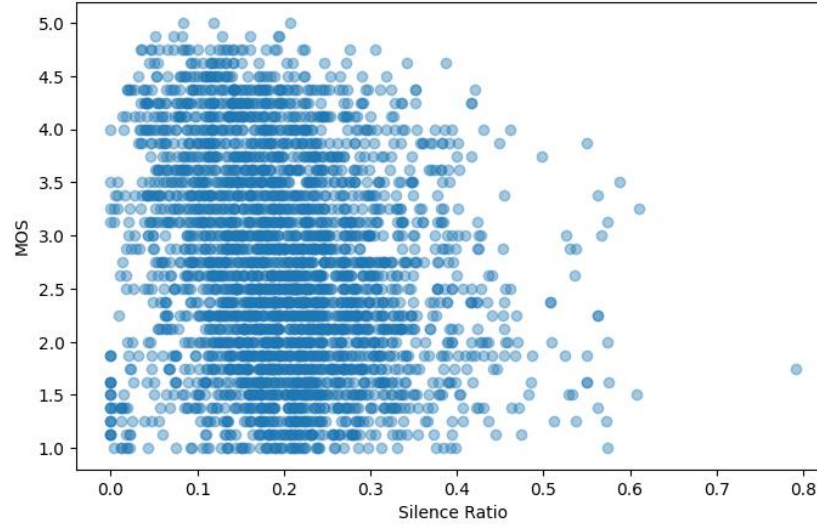


Figure 4. Silence ratio vs. MOS.

3.3 RQ2/RQ3: Joint Predictive Model

A multiple linear regression combining all 15 acoustic features was trained to predict MOS. On the held-out test set, the model achieved the performance shown in Table 3.

Metric	Value
R^2	0.184
MAE	0.717
RMSE	0.869

Table 3. Linear regression performance on the held-out test set.

An R^2 of 0.184 indicates the model explains roughly 18% of the variance in MOS scores — acoustic features alone are only a weak-to-moderate predictor of perceived naturalness. Figure 5 shows the standardized regression coefficients (feature importance) for each feature.

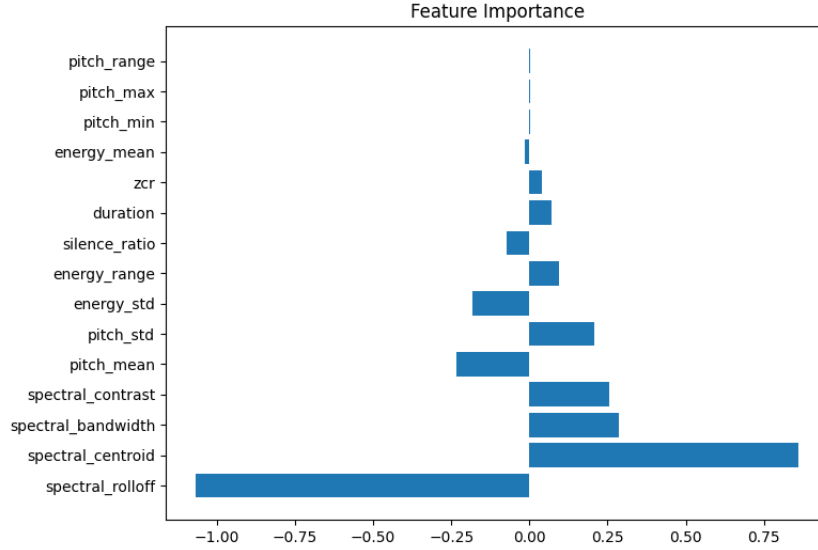


Figure 5. Standardized linear regression coefficients by feature.

Once all features are considered jointly, spectral features dominate the model: spectral rolloff (-1.067) and spectral centroid ($+0.860$) have by far the largest standardized effects, followed by spectral bandwidth and spectral contrast. This is notable because `spectral_contrast` alone had the strongest simple correlation, but in the joint model `spectral_rolloff` and `spectral_centroid` carry more predictive weight — suggesting these features interact with or partially capture similar information. Pitch-range statistics (min, max, range) contribute almost nothing to the joint model once other features are accounted for.

3.4 RQ3: High- vs. Low-Naturalness Characteristics

Taken together, the correlation and regression results suggest that TTS systems rated as more natural-sounding tend to exhibit lower spectral rolloff and less silence, alongside higher spectral centroid and spectral contrast. Pitch-related statistics (mean, std, range) show comparatively weak, near-negligible differences between higher- and lower-scoring systems in this dataset, suggesting that in this dataset, spectral shape and pacing (silence) distinguish natural from unnatural-sounding TTS output more than pitch variability does.

4. Discussion

The results partially answer the project's research questions: acoustic features do relate to naturalness, but no single feature is strongly predictive on its own, and even a full linear combination of all extracted features explains less than a fifth of the variance in human MOS ratings. This points to naturalness being a multidimensional perceptual judgment that likely depends on prosodic patterns, phoneme-level artifacts, and other qualities not fully captured by whole-clip summary statistics such as means and standard deviations.

The relative importance of spectral features (rolloff, centroid, contrast) over pitch and energy statistics suggests that the overall 'brightness' or spectral shape of synthesized speech may be a more consistent naturalness signal than pitch variability across this dataset. This is a useful, if modest, insight for prioritizing which acoustic properties to monitor when evaluating or improving TTS systems.

4.1 Limitations

- R^2 of 0.184 indicates acoustic features explain only a modest share of MOS variance; non-linear or higher-order relationships, and non-acoustic (e.g. linguistic/semantic) factors, are not captured.

- All features are simple aggregate statistics (mean/std/min/max/range) over an entire clip, which may smooth over locally important acoustic events.
- Only a linear model was tested; non-linear models were not explored in this analysis.
- The analysis is at the individual-utterance level rather than aggregated per TTS system, which limits direct system-to-system comparison.

4.2 Future Work

- Evaluate non-linear models (e.g. Random Forest, Gradient Boosting) to test whether they capture more MOS variance and to cross-check feature importance rankings.
- Aggregate results per TTS system to enable direct system-level comparisons of acoustic profiles, as originally outlined in the project's research plan.
- Explore interaction effects between spectral and pitch features.
- Incorporate additional speech-quality descriptors from the literature, such as jitter, shimmer, and formant-based measures.

5. Conclusion

This project extracted acoustic features from 3,382 TTS audio samples and analyzed their relationship with human-rated naturalness (MOS). Spectral contrast and silence ratio emerged as the strongest individual correlates of naturalness, while spectral rolloff and spectral centroid dominated a joint linear regression model that explained approximately 18% of MOS variance. These findings suggest that spectral shape and speech pacing are more consistently associated with perceived TTS naturalness than pitch variability in this dataset, while also highlighting that a substantial portion of what makes synthetic speech sound natural remains unexplained by simple acoustic summary statistics alone.

References

VoiceMOS Challenge Dataset. Retrieved from Zenodo: <https://zenodo.org/records/10691660>

McFee, B., et al. librosa: Audio and Music Signal Analysis in Python: <https://proceedings.scipy.org/articles/Majora-7b98e3ed-003.pdf>

Project repository: <https://github.com/sassdahRAK/Acoustic-TTS-Analysis>